Farm Animals – Java Classes and Objects

1. Objective

To develop a Java program using classes and objects to represent farm animals and demonstrate constructors, methods, parameter passing, and method overloading.

2. Task Description

Create an Animal class containing animal name, place, food, and sound. Use a constructor to initialize these values. Create methods to display animal details, accept food as a parameter, and demonstrate method overloading. In the main class, create objects for Cow, Pig, and Horse and display their details.

3. Step-wise Execution

1. Create the Animal class and declare name, place, food, and sound variables.

2. Create a parameterized constructor to initialize the animal details.

3. Create the display() method to print all animal details.

4. Create the eat(String foodName) method to demonstrate parameter passing.

5. Create the sound(String s) method to demonstrate an overloaded method name.

6. Create Animal objects for Cow, Pig, and Horse in the FarmAnimals main class.

7. Call display() for each object and print the details.

8. Call eat() with appropriate food values.

9. Call sound() for the Cow object with a sound value.

10. Run the program and observe the output.

4. Algorithm

Step 1: Start.

Step 2: Define the Animal class with four data members.

Step 3: Define a constructor to initialize the data members.

Step 4: Define display(), eat(), and sound() methods.

Step 5: Create three Animal objects: Cow, Pig, and Horse.

Step 6: Display the details of all three objects.

Step 7: Pass food values to the eat() method.

Step 8: Pass a sound value to the sound() method.

Step 9: Stop.

5. UML Diagram

Animal	
Data Members	**Methods**
name : String	Animal(String n, String p, String f, String s)
place : String	display() : void

food : String	eat(String foodName) : void
sound : String	sound(String s) : void

Objects created in main(): cow, pig, horse → Animal

6. Program Code

```
class Animal {
    String name;
    String place;
    String food;
    String sound;

    // Constructor
    Animal(String n, String p, String f, String s) {
        name = n;
        place = p;
        food = f;
        sound = s;
    }

    // Method
    void display() {
        System.out.println("Animal: " + name);
        System.out.println("Stays in: " + place);
        System.out.println("Eats: " + food);
        System.out.println("Sound: " + sound);
    }

    // Method with parameter
    void eat(String foodName) {
        System.out.println(name + " eats " + foodName);
    }

    // Method overloading
    void sound(String s) {
        System.out.println(name + " says " + s);
    }
}

public class FarmAnimals {
    public static void main(String[] args) {

        // Objects
        Animal cow = new Animal("Cow", "Cowshed", "Grass", "Moo");
        Animal pig = new Animal("Pig", "Pigsty", "Corn", "Oink");
        Animal horse = new Animal("Horse", "Stable", "Grass", "Neigh");

        // Display details
        cow.display();
        System.out.println();

        pig.display();
        System.out.println();

        horse.display();
        System.out.println();

        // Parameter passing
        cow.eat("Grass");
        pig.eat("Corn");
        horse.eat("Grass");

        // Method overloading
        cow.sound("Moo Moo");
    }
}
```

7. Output Screen

```
Animal: Cow
Stays in: Cowshed
Eats: Grass
Sound: Moo

Animal: Pig
Stays in: Pigsty
Eats: Corn
Sound: Oink

Animal: Horse
Stays in: Stable
Eats: Grass
Sound: Neigh

Cow eats Grass
Pig eats Corn
Horse eats Grass
Cow says Moo Moo
```

8. Result

The Farm Animals Java program was executed successfully. It demonstrates classes and objects, a constructor, methods with parameters, and the sound(String) method using the given program.